

Experiment No: 5

Aim: Implement the embedding wallet (Metamask) and transaction using Solidity

Theory:

1. Blockchain

Blockchain is a decentralized and distributed digital ledger used to store transactions securely. Instead of storing data on a single central server, the information is maintained across a network of computers. Each transaction is recorded in a block and connected to previous blocks, making the data difficult to modify.

2. Cryptocurrency Wallet

A cryptocurrency wallet is a software application that allows users to manage their blockchain accounts and interact with decentralized applications. It stores and manages the user's private keys and provides an interface for sending transactions and interacting with smart contracts.

3. MetaMask

MetaMask is a cryptocurrency wallet that acts as a bridge between users, blockchain networks, and decentralized applications (DApps). It allows users to create or import blockchain accounts, manage cryptocurrency, connect to networks such as Ethereum and Sepolia, and approve blockchain transactions.

In this experiment, MetaMask is used as the wallet to connect Remix IDE with the Sepolia test network.

4. Testnet

A testnet is a blockchain network used for development and testing. It behaves similarly to a real blockchain but uses test cryptocurrency that has no real monetary value. Developers can use testnets to deploy and test smart contracts without spending real cryptocurrency.

Sepolia is an Ethereum test network used for testing smart contracts and blockchain applications. Sepolia ETH is used to pay transaction gas fees on the test network.

5. Solidity

Solidity is a programming language used to develop smart contracts for Ethereum-compatible blockchain networks. It supports variables, functions, addresses, constructors, and other programming features required to create decentralized applications.

6. Smart Contract

A smart contract is a program stored and executed on a blockchain. It automatically performs predefined operations when its functions are called. Once deployed, the contract receives a unique blockchain address and can be interacted with through a wallet or DApp.

In this experiment, a simple Solidity smart contract is created to store and update a message.

7. Remix IDE

Remix is a web-based development environment used to write, compile, deploy, and interact with Solidity smart contracts. It provides tools for compiling Solidity code and connecting to blockchain wallets such as MetaMask.

In this experiment, Remix is connected to MetaMask using the **Browser Extension** environment.

8. Blockchain Transaction

A blockchain transaction is an operation submitted to the blockchain. Transactions may involve transferring cryptocurrency, deploying a smart contract, or calling a function that changes the contract's stored data.

For example, when the `setMessage()` function is called, a transaction is created because the stored value of the smart contract is changed. MetaMask asks the user to confirm the transaction and pay the required gas fee.

9. Gas Fee

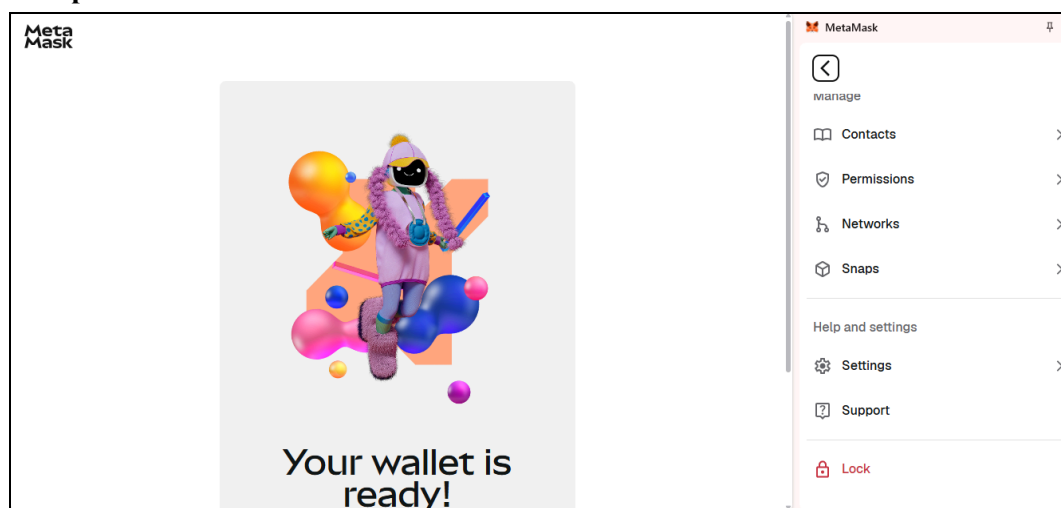
Gas is the computational work required to execute operations on an Ethereum-compatible blockchain. A gas fee is paid for processing a transaction. On the Sepolia testnet, the gas fee is paid using test ETH.

10. Etherscan

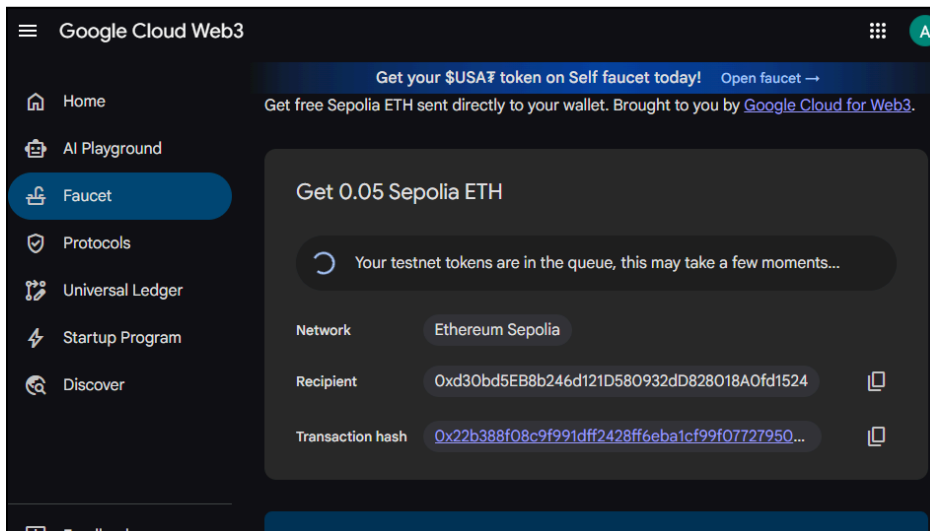
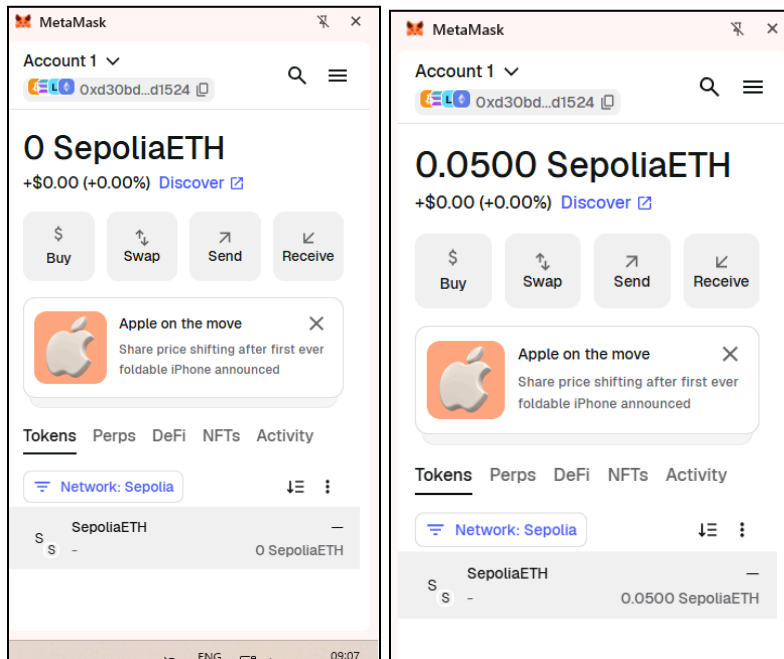
Etherscan is a blockchain explorer used to view transactions, blocks, wallet addresses, smart contracts, and other blockchain information. In this experiment, the Sepolia version of Etherscan is used to verify the transaction and check details such as transaction hash, status, sender, receiver, block number, and gas used.

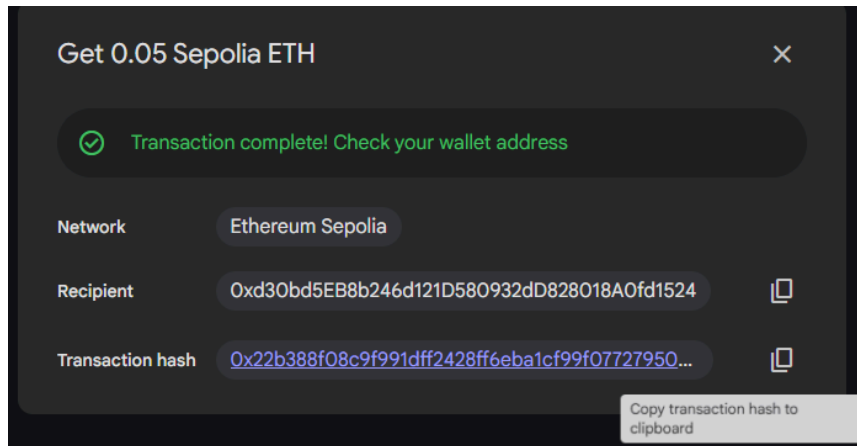
Output:

Set Up MetaMask

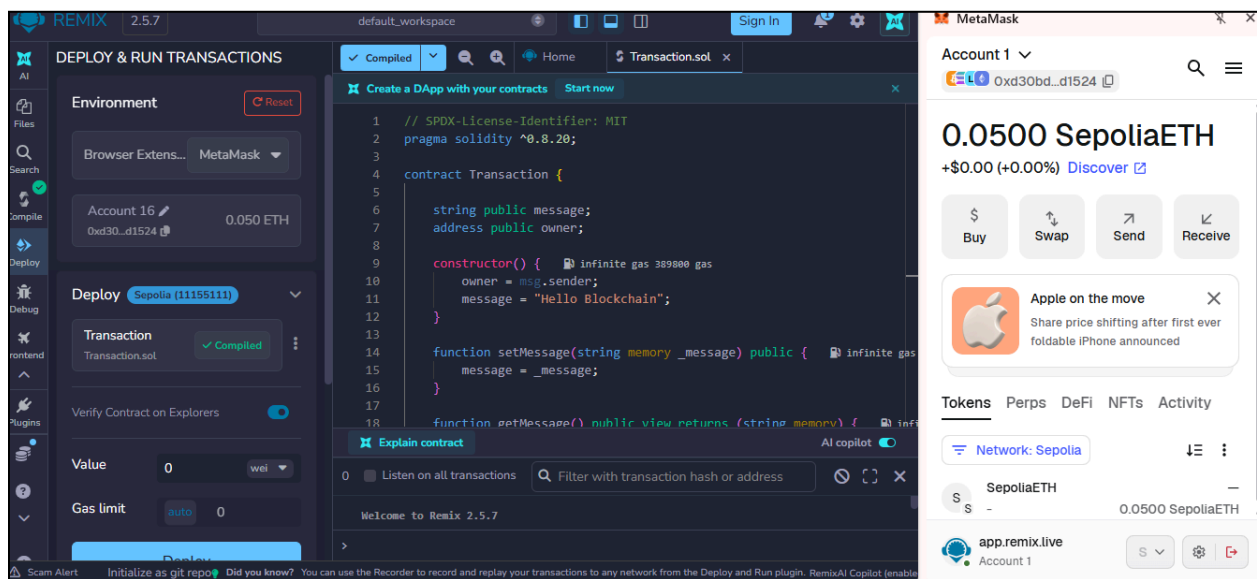


Connect Sepolia Testnet and transfer free eth

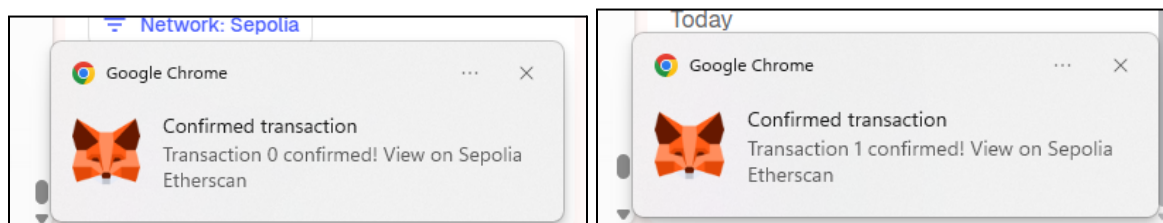




Connect MetaMask with Remix IDE and Create Solidity Smart Contract



Compile and deploy the Contract



Contract interaction

Status

Confirmed

Date

Sep 10, 2026, 9:14 AM

Account

Account 1 (0xd30bd...d1524)

Network

S Sepolia

Transaction ID

0x738a4...7a2f3

Account 1

Transaction request

Estimated changes

Network

s Sepolia

Request from

app.remix.live

Interacting with

0xFE9B0...E912C

Network fee

Speed

Move notification to Notification Center

Google Chrome

Confirmed transaction
Transaction 2 confirmed! View on Sepolia Etherscan

Verify Transaction on Sepolia Etherscan

TRANSACTION ACTION

Call

Redeem Delegations

Function by 0xd30bd5EB...8A0fd1524 on 0xdb9B1e94...141047dB3

[This is a Sepolia Testnet transaction only]

Transaction Hash:

0xd7e9956c047b77e1fb07f3d3c1040903aa6253af4da3439f649f2296bbe20d0c

Status:

Success

Block:

11672431

4 Block Confirmations

Timestamp:

51 secs ago

Sep-10-2026 03:53:00 AM +UTC

From:

0xd30bd5EB8b246d121D580932dD828018A0fd1524

To:

0xdb9B1e94B5b69Df7e401DDbedE43491141047dB3

Value:

0 ETH

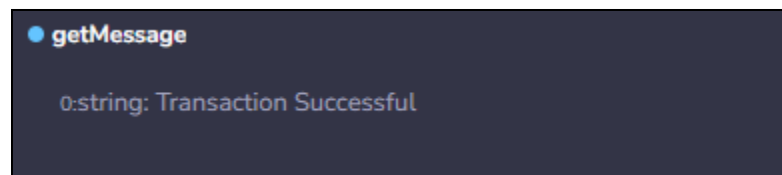
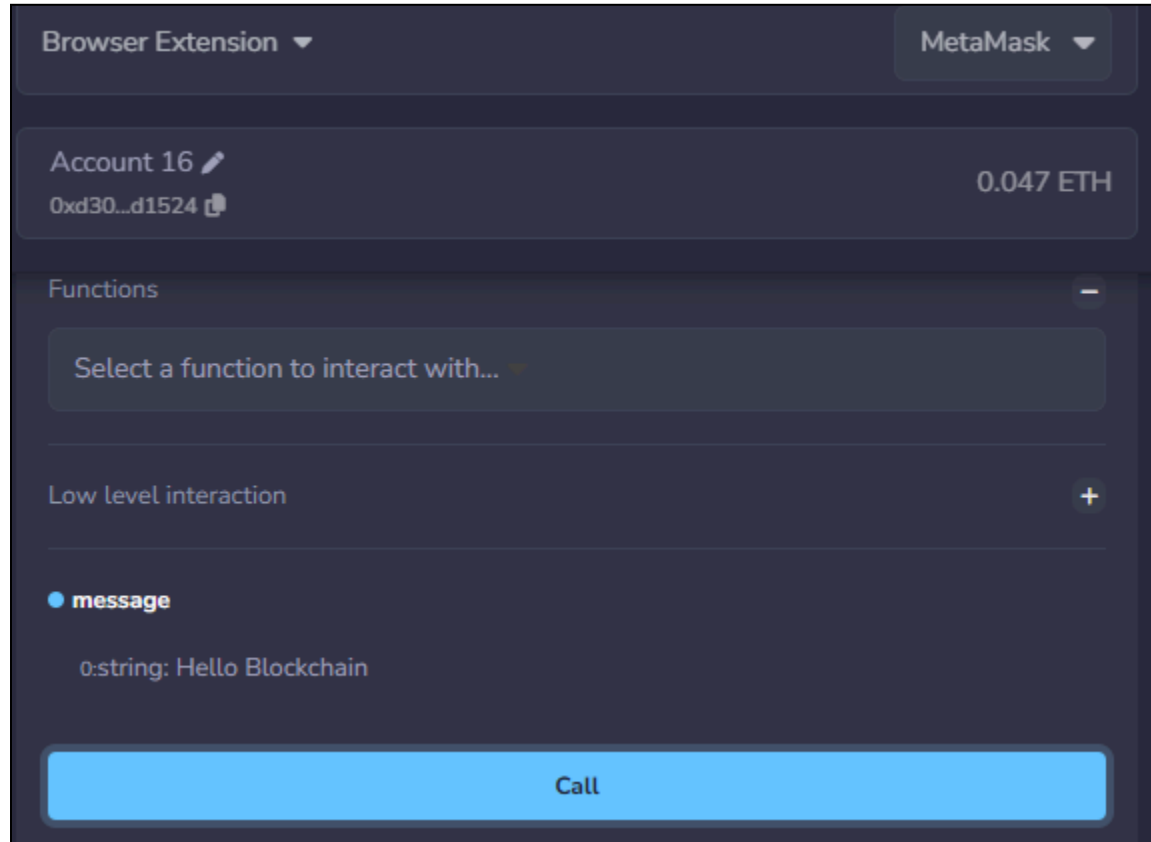
Transaction Fee:

0.000283042487404716 ETH

Gas Price:

2.465892052 Gwei (0.000000002465892052 ETH)

Interact with the Smart Contract



Conclusion:

This experiment helped me understand how a cryptocurrency wallet such as MetaMask can be integrated with Remix IDE to interact with a blockchain. I learned how to create and deploy a Solidity smart contract and perform transactions using a MetaMask account. The experiment also demonstrated the role of testnets in safely testing blockchain applications without using real cryptocurrency.